Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-G Congenital Heart Conditions

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-07-C01-BD1

Brain dump 1 · foundational retrieval: Draw the three left-to-right shunts and label the pressures

A competent response contains

· After birth the left ventricle runs at about 120/10 mm Hg and the right ventricle at about 25/5 mm Hg, so a VSD, an ASD or a PDA sends oxygenated blood back toward the lungs.

· Blood crosses a defect toward the side with lower pressure, or, between filling chambers, toward the side that fills more easily.

· VSD and PDA are driven by the systolic and diastolic pressure difference; an ASD is driven mainly by the greater compliance of the thin-walled right ventricle.

· Qp:Qs is the ratio of pulmonary blood flow to systemic blood flow; above 1 means a net left-to-right shunt and below 1 a net right-to-left shunt.

· Systemic arterial blood stays fully saturated in all three because no desaturated blood reaches the aorta, so the child is not cyanotic.

· The cost of a left-to-right shunt is volume: the lungs and the chambers carrying the recirculated blood receive more flow than the body uses.

· A defect before the tricuspid valve (ASD) adds flow at low pressure; a defect after it (VSD, PDA) transmits systemic pressure as well as extra flow, so pulmonary vascular disease develops earlier and more often with VSD and PDA.

Distinctions that matter

· Pressure drives the ventricular and ductal shunts; compliance drives the atrial shunt, which is why atrial pressures only a few mm Hg apart still produce a large shunt.

· Extra flow and transmitted pressure are separate insults to the pulmonary arteries, and where the defect sits relative to the tricuspid valve decides which the lungs receive.

Common misconceptions

· Calling any shunt a cause of cyanosis. A left-to-right shunt leaves systemic saturation normal because the extra blood goes to the lungs, not to the body.

· Describing the ASD shunt as pressure driven. Left and right atrial pressures differ by only a few mm Hg; ventricular compliance is what moves the blood.

Where it is taught: Section C01, first two paragraphs and the shunt categories table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-BD2

Brain dump 2 · integration and application: Explain every Eisenmenger finding from the reversal

A competent response contains

· Chronic high flow and pressure injure the pulmonary endothelium: nitric oxide and prostacyclin fall and endothelin-1 rises.

· The arterioles show medial hypertrophy, then intimal proliferation and fibrosis, and finally plexiform lesions, the same changes as in pulmonary arterial hypertension.

· As PVR rises the right ventricle hypertrophies, the shunt shrinks, becomes bidirectional, and finally reverses when right-sided pressure exceeds left-sided pressure.

· Cyanosis appears late because it begins only once desaturated blood reaches the systemic arteries.

· Oxygen barely helps because the shunted blood never passes the alveoli.

· Renal hypoxia raises erythropoietin, giving secondary polycythemia, which raises viscosity and makes the cyanosis more visible.

· Clubbing accompanies chronic hypoxemia; a leading explanation is that platelet clumps bypass the pulmonary capillaries and release growth factors in the fingertips.

· A loud P2 reflects raised pulmonary arterial pressure, and the original murmur softens because the pressure difference across the defect has disappeared.

· An ASD of the same size adds flow at low pressure, so pulmonary vascular disease develops later and in a minority of patients.

· Advanced remodeling, once plexiform lesions have formed, is irreversible.

Distinctions that matter

· A softening murmur in this setting signals worsening disease rather than improvement, because murmur loudness tracks the gradient across the defect.

· Polycythemia here is secondary and appropriate to hypoxia, not a primary marrow disorder.

Common misconceptions

· Expecting supplemental oxygen to correct the cyanosis. The shunted blood bypasses the alveoli entirely.

· Treating the reversal as the start of the disease. The reversal is the end stage of years of pulmonary vascular remodeling.

Where it is taught: Section C01, third and fifth paragraphs and the flow diagram of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-RT01

Retrieval exercise: Rebuild the path from shunt to reversal

A competent response contains

· Step 1: a left-to-right shunt overloads the pulmonary circulation.

· Step 2: high flow and pressure injure the pulmonary endothelium.

· Step 3: medial hypertrophy, intimal fibrosis and plexiform lesions raise PVR.

· Step 4: the right ventricle hypertrophies as its pressure rises.

· Step 5: right-sided pressure exceeds left-sided pressure and the shunt reverses.

· Step 6: late cyanosis, clubbing and polycythemia appear.

· Plexiform lesions mark advanced, irreversible remodeling.

· Reversal requires right-sided pressure to exceed left-sided pressure, so the arrow moves left to right early, both ways at the midpoint, and right to left at the end.

· Systemic arterial saturation is normal early, mildly reduced when the shunt is bidirectional, and clearly low after reversal.

Distinctions that matter

· Resistance rises before the shunt direction changes, so a shrinking shunt is an early sign of the same process that later reverses it.

Common misconceptions

· Drawing a sudden switch from left-to-right to right-to-left. The shunt passes through a bidirectional stage as pressures equalize.

Where it is taught: Section C01, flow diagram on Eisenmenger syndrome of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C01-PR01

Prediction exercise: Predict what happens as pulmonary vascular resistance falls after birth

A competent response contains

· Shunt volume increases, because falling PVR lowers the resistance to pulmonary flow while systemic resistance is unchanged.

· Qp:Qs rises further above 1 as pulmonary flow climbs relative to systemic flow.

· Pulmonary blood flow increases, and the recirculated volume returns through the left atrium and left ventricle, so both dilate.

· In a large nonrestrictive defect the murmur is soft or short and does not become loud, because ventricular pressures are nearly equal and there is little gradient to generate turbulence; an apical mid-diastolic rumble from high mitral flow and a loud P2 appear instead.

· By the second month the infant shows heart failure: tachypnea, sweating and fatigue with feeds, poor weight gain, and hepatomegaly.

· Systemic arterial saturation stays normal, because the shunt is left to right and no desaturated blood reaches the aorta.

Distinctions that matter

· Falling PVR explains why a large shunt causes few symptoms at birth and heart failure within one to two months.

· Volume overload and hypoxemia are different burdens; this infant has the first and not the second.

Common misconceptions

· Expecting a large defect to produce a loud murmur. Loudness follows the gradient, so the loudest defects are often the smallest.

· Expecting cyanosis as the shunt grows. More flow to the lungs does not desaturate systemic blood.

Where it is taught: Section C01, second paragraph; Section C02, final paragraph and the small versus large VSD table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-BD1

Brain dump 1 · foundational retrieval: Build the ventricular septal defect from the septum outward

A competent response contains

· The interventricular septum has a thick muscular part and a small membranous part sitting just below the aortic valve.

· The membranous septum forms last from fusion of the endocardial cushions, the conotruncal ridges, and the muscular septum, so it is most often left incomplete.

· Most VSDs are membranous or perimembranous; muscular defects lie within the trabeculated septum and may be multiple.

· Left ventricular pressure exceeds right from the start of isovolumetric contraction to the end of ejection, so blood crosses throughout systole.

· The murmur is harsh and holosystolic, starts with S1 and often obscures it, is loudest at the left lower sternal border in the third and fourth intercostal spaces, and frequently comes with a thrill.

· Sustained handgrip raises systemic resistance, increases the left-to-right gradient, and makes the murmur louder.

· Shunted blood leaves the left ventricle in systole and passes almost directly into the pulmonary artery, so the volume load falls on the pulmonary circulation, the left atrium and the left ventricle.

· The right ventricle enlarges mainly when pulmonary pressure rises, not from the shunt volume itself.

· Associations: trisomy 21, fetal alcohol exposure, other chromosomal syndromes, and VSD is one of the four components of tetralogy of Fallot.

· VSD is the most common congenital heart defect at birth when bicuspid aortic valve is not counted.

Distinctions that matter

· A VSD loads the left atrium and left ventricle even though the defect lies between the ventricles, because the shunted blood is routed through the lungs and back to the left heart.

· Membranous and muscular defects differ in location and in how often they close, not in the direction of the shunt.

Common misconceptions

· Expecting right ventricular enlargement from the shunt volume. The blood crosses during systole and leaves through the pulmonary artery rather than filling the right ventricle.

· Placing the membranous septum away from the aortic valve. It sits just below it, which is why a cusp can prolapse into a defect there.

Where it is taught: Section C02, first three paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-BD2

Brain dump 2 · integration and application: Two infants, two murmurs, two courses

A competent response contains

· Infant A has the smaller restrictive defect: a large pressure gradient persists across it, right ventricular pressure stays low, and the high-velocity jet makes a loud harsh holosystolic murmur with a thrill.

· Infant B has the larger nonrestrictive defect: the gradient is small, right ventricular pressure approaches left ventricular pressure, so the murmur is softer and shorter, with an apical mid-diastolic rumble from high mitral flow and a loud P2.

· In a small defect the shunt size is limited by the defect itself; in a large defect ventricular pressures equalize and the shunt is set by the ratio of pulmonary to systemic resistance.

· Infant A's defect, especially a muscular one, often closes spontaneously in early childhood as surrounding muscle grows or as tricuspid septal leaflet tissue covers a perimembranous opening; endocarditis remains a risk if it persists.

· Infant B has the findings of heart failure as PVR falls, and without correction pulmonary vascular disease can become irreversible within the first few years, followed by Eisenmenger reversal.

· A defect just below the aortic valve can let an aortic cusp prolapse into it, adding aortic regurgitation.

· The high-velocity jet damages the right ventricular endocardium where it strikes, creating a site for endocarditis.

Distinctions that matter

· Murmur loudness reports the pressure gradient, not the size of the hole, so the loudest VSDs are often the smallest.

· Spontaneous closure belongs to small defects; irreversible pulmonary vascular disease belongs to large ones.

Common misconceptions

· Reading a soft murmur as a small defect. A soft murmur with a loud P2 and failure to thrive points to a large defect with nearly equal ventricular pressures.

· Assuming the mid-diastolic rumble means mitral stenosis. It comes from the large volume crossing a normal mitral valve.

Where it is taught: Section C02, small versus large VSD table and final paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-BD1

Brain dump 1 · foundational retrieval: Build the atrial septum, then explain the fixed split S2

A competent response contains

· The septum primum grows toward the endocardial cushions and closes the ostium primum, while perforations in its upper part create the ostium secundum.

· The septum secundum then grows on the right side of the septum primum, leaving the foramen ovale, whose flap valve is the septum primum.

· In the fetus high right atrial pressure pushes the flap open; after birth left atrial pressure rises above right and presses it shut, and the two septa usually fuse over the following months.

· An ASD shunts left to right mainly because the thin-walled right ventricle is more compliant and fills more easily than the left, not because of a large pressure difference.

· The extra volume passes through the right atrium, right ventricle and pulmonary arteries and enlarges them, while the left atrium decompresses through the defect.

· Flow across the defect itself is low velocity and silent.

· The systolic ejection murmur comes from increased flow across a normal pulmonic valve and is heard at the left upper sternal border; a large shunt adds a mid-diastolic rumble from increased tricuspid flow.

· Normally inspiration increases right heart venous return and delays pulmonic closure, so S2 splits in inspiration and narrows in expiration.

· In an ASD the right ventricle already carries a large stroke volume in both phases: in inspiration the extra systemic venous return is offset by less left-to-right shunting, and in expiration the shunt grows again.

· Right ventricular volume, and therefore the delay of P2, stays nearly constant, giving a wide fixed split S2.

Distinctions that matter

· The murmur of an ASD is a pulmonic flow murmur, not the sound of blood crossing the defect.

· Wide splitting and fixed splitting are different statements: the width comes from the large right ventricular stroke volume, the fixity from the offsetting of venous return and shunt flow.

Common misconceptions

· Explaining the left-to-right atrial shunt by atrial pressure alone. Ventricular compliance is the main driver.

· Expecting the defect itself to be audible. Velocity across it is too low to generate turbulence.

Where it is taught: Section C03, first paragraph and the paragraphs on shunt direction and the split S2 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-BD2

Brain dump 2 · integration and application: A stroke at 34 with no murmur

A competent response contains

· A patent foramen ovale explains the stroke: the septum primum and septum secundum never fused, so the flap is unsealed although no tissue is missing.

· While left atrial pressure exceeds right, the flap is held against the septum secundum, no blood crosses, and there is neither a murmur nor volume overload.

· Transient reversal occurs when right atrial pressure exceeds left, as on release of a Valsalva maneuver, with coughing or straining, or with acute pulmonary embolism.

· A deep vein thrombus can then cross to the left atrium and reach the systemic arteries, causing stroke or limb ischemia; this is a paradoxical embolism.

· A patent foramen ovale is present in about one in four adults, which is why an unexplained stroke in a young adult raises suspicion of a PFO or ASD.

· Adult presentations of an unrecognized ASD: atrial fibrillation or flutter, exercise intolerance, and right heart failure, with Eisenmenger syndrome in a minority.

· The rhythm problem follows right atrial stretch from years of extra volume.

· An ASD is missing septal tissue; a patent foramen ovale is complete tissue that never sealed.

· An ostium primum ASD is the lesion with a cleft anterior mitral leaflet and mitral regurgitation, and it is associated with trisomy 21.

Distinctions that matter

· A patent foramen ovale allows embolism without any resting shunt, while an ASD carries a continuous shunt and enlarges the right heart.

· Secundum and primum defects are both tissue deficits; only the primum defect involves the endocardial cushions and the mitral valve.

Common misconceptions

· Assuming any intracardiac communication must cause a murmur. A PFO is silent because nothing crosses it at rest.

· Expecting an ASD to declare itself in childhood. Many are found only in adulthood, often through an arrhythmia.

Where it is taught: Section C03, atrial septal lesions table, final paragraph, and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-CP01

Comparison table: The three lesions of the atrial septum

A competent response contains

· Secundum ASD: excess resorption of the septum primum or a septum secundum too small to cover the ostium secundum; a true tissue deficit at the fossa ovalis; the most common ASD; fetal alcohol exposure and Holt-Oram syndrome with upper limb defects; no specific valve lesion; ECG right axis deviation with incomplete right bundle branch block; a resting shunt large enough to overload the right heart.

· Primum ASD: the septum primum fails to meet the endocardial cushions, making a partial atrioventricular septal defect; a tissue deficit; associated with trisomy 21; usually a cleft anterior mitral leaflet with mitral regurgitation; ECG left or superior axis deviation; a resting shunt is present.

· Patent foramen ovale: the septum primum and septum secundum fail to fuse, so no tissue is missing and the flap is simply unsealed; present in about one in four adults; no specific syndrome association; no valve lesion; no characteristic ECG change; no resting shunt while left atrial pressure exceeds right.

· Axis direction is the quick separator between the two ASDs: right axis deviation for secundum, left or superior axis deviation for primum.

Distinctions that matter

· Missing tissue against unsealed tissue separates an ASD from a patent foramen ovale and explains why only one of them overloads the right heart.

· The primum defect is an endocardial cushion lesion, which is why it carries a mitral valve abnormality and the secundum defect does not.

Common misconceptions

· Treating a patent foramen ovale as a small ASD. It is unfused complete tissue with no resting shunt.

· Assigning right axis deviation to both ASDs. The primum defect gives a left or superior axis.

Where it is taught: Section C03, table of atrial septal lesions of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-BD1

Brain dump 1 · foundational retrieval: The ductus arteriosus from arch to ligament

A competent response contains

· The ductus arteriosus develops from the distal part of the left sixth aortic arch.

· It joins the pulmonary trunk near its bifurcation, at the root of the left pulmonary artery, to the aortic arch just distal to the origin of the left subclavian artery.

· In the fetus pulmonary vascular resistance is high, so most right ventricular output bypasses the lungs through the ductus into the descending aorta.

· Its smooth muscle is held relaxed by low fetal oxygen tension and by prostaglandin E2, produced by the placenta and the ductus and poorly cleared by the unventilated fetal lungs.

· Closure sequence: first breaths expand the lungs and PVR falls; arterial PO2 rises and constricts ductal smooth muscle; the placenta is removed and the perfused lungs clear PGE2; the ductus closes functionally within the first few days; intimal proliferation and fibrosis form the ligamentum arteriosum.

· Prematurity is the major risk factor, because immature ductal muscle constricts weakly to oxygen and stays sensitive to PGE2.

· Congenital rubella, especially first trimester infection, causes PDA with cataracts and sensorineural deafness.

· Birth at high altitude, where arterial PO2 is low, also predisposes.

· The left recurrent laryngeal nerve hooks under the ligamentum arteriosum, which is why surgery in this region can cause hoarseness.

Distinctions that matter

· Oxygen and prostaglandin are two separate closure signals, and a ductus that misses either one can stay open.

· Functional closure in the first days is muscular; anatomic closure forming the ligamentum arteriosum takes weeks to months.

Common misconceptions

· Placing the ductus proximal to the left subclavian artery. It joins the arch just distal to it, which is what produces differential cyanosis later.

· Naming the sixth arch generally. It is the distal part of the left sixth arch.

Where it is taught: Section C04, first paragraph, closure flow diagram, and the risk factor paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-BD2

Brain dump 2 · integration and application: Pulses, pressures and blue toes

A competent response contains

· After birth aortic pressure exceeds pulmonary arterial pressure in both systole and diastole, so flow runs from aorta to pulmonary artery throughout the cycle, giving a continuous machine-like murmur loudest below the left clavicle and peaking around S2.

· Diastolic pressure falls because the aorta drains into the low-resistance pulmonary circuit during diastole.

· Systolic pressure rises because the extra left ventricular stroke volume is larger, so the pulse pressure widens and the pulses are bounding.

· The recirculated blood returns to the left atrium and left ventricle, so both dilate, and a large PDA causes heart failure.

· In a premature infant whose PVR is still high, the diastolic gradient may be small and the murmur only systolic.

· With Eisenmenger physiology, desaturated pulmonary arterial blood enters the aorta beyond the left subclavian artery, so the lower body receives desaturated blood while the head and arms are supplied from vessels proximal to the ductus.

· The result is differential cyanosis: cyanotic clubbed toes with pink fingers.

· Other complications are infective endarteritis at the ductal jet and, rarely, ductal aneurysm.

Distinctions that matter

· The continuous murmur reflects a gradient in both systole and diastole, which is why a premature infant with high PVR may have only a systolic murmur.

· The wide pulse pressure has two separate causes, diastolic runoff and increased stroke volume, and both should be stated.

Common misconceptions

· Expecting cyanosis in the arms with a reversed PDA. The arch vessels arise proximal to the ductus and carry undiluted left ventricular blood.

· Treating a continuous murmur as evidence of severity. Its presence reports the gradient, and losing it can mean rising pulmonary pressure.

Where it is taught: Section C04, fourth and fifth paragraphs and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-RT01

Retrieval exercise: Rebuild ductal closure after birth

A competent response contains

· Step 1: the first breaths expand the lungs and PVR falls.

· Step 2: arterial PO2 rises and constricts ductal smooth muscle.

· Step 3: the placenta is removed and the perfused lungs clear PGE2.

· Step 4: the ductus closes functionally within the first few days.

· Step 5: intimal proliferation and fibrosis form the ligamentum arteriosum, with anatomic closure taking weeks to months.

· PGE2 is the prostaglandin that falls, and it falls because the placenta is gone and the now perfused lungs clear it.

· Cyclooxygenase inhibition lowers PGE2 and promotes closure; exogenous prostaglandin E1 holds the ductus open.

· The premature ductus is at highest risk because its immature muscle constricts weakly to oxygen and it remains sensitive to PGE2.

Distinctions that matter

· Pharmacologic opening and closing confirm which signal is being manipulated: prostaglandin keeps it open, and blocking prostaglandin synthesis closes it.

Common misconceptions

· Attributing closure to oxygen alone. Rising oxygen and falling PGE2 act together.

Where it is taught: Section C04, flow diagram on ductal closure and the following paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-BD1

Brain dump 1 · foundational retrieval: Build tetralogy of Fallot from one developmental error

A competent response contains

· One error explains all four features: anterosuperior and leftward displacement of the infundibular (outlet) septum.

· The displaced septum crowds the right ventricular outflow tract, producing infundibular pulmonary stenosis, often with a small pulmonary valve and arteries.

· Because it no longer lines up with the muscular septum, it leaves a large malalignment VSD.

· The aortic root, shifted rightward, sits above that defect, giving the overriding aorta.

· Right ventricular hypertrophy is the consequence rather than a primary defect, because the right ventricle pumps against the obstruction at systemic pressure through the unrestrictive VSD.

· Neural crest cells migrate into the conotruncal ridges and help build the outflow septum.

· That explains the link with 22q11.2 deletion (DiGeorge syndrome), which also causes thymic and parathyroid hypoplasia and is associated with truncus arteriosus and interrupted aortic arch.

· Other associations are trisomy 21 and fetal alcohol exposure.

· Tetralogy of Fallot is the most common cyanotic congenital heart defect.

Distinctions that matter

· Three features are structural results of the displaced septum; the hypertrophy is a response to the pressure load, so it is the one that is secondary.

· Conotruncal lesions share a neural crest origin, which is why tetralogy, truncus arteriosus and interrupted arch travel with the same deletion.

Common misconceptions

· Listing the four features as four separate developmental mistakes. They follow from a single displacement of the outlet septum.

· Calling the right ventricular hypertrophy a congenital malformation. It develops because the right ventricle works at systemic pressure against the obstruction.

Where it is taught: Section C05, first and second paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-BD2

Brain dump 2 · integration and application: Why one child with tetralogy is blue and another is not

A competent response contains

· The variable is the severity of right ventricular outflow obstruction, which sets how much blood takes the aortic exit.

· With a large VSD the ventricles share one pressure, so flow divides between the stenotic pulmonary outflow and the VSD into the aorta in inverse proportion to their resistances, the aortic route carrying the resistance of the systemic circulation.

· Severe stenosis diverts deoxygenated blood across the VSD into the aorta and causes early cyanosis; mild stenosis gives a balanced or even left-to-right shunt and little cyanosis.

· The VSD is silent because there is no pressure gradient across it.

· The murmur is a systolic ejection murmur at the left upper sternal border from flow through the narrowed outflow tract, and S2 is single because P2 is soft.

· Pulmonary blood flow is reduced, so pulmonary vascular remodeling and Eisenmenger physiology do not develop.

· Spell loop: crying, feeding or dehydration narrows the outflow or lowers SVR; more right-to-left flow crosses the VSD; arterial PO2 falls and metabolic acidosis develops; hyperpnea increases systemic venous return to the right ventricle; acidosis lowers SVR further and the shunt increases again.

· The murmur softens or disappears during a spell because less blood crosses the obstructed outflow tract.

· Squatting kinks the femoral arteries and raises SVR, so more right ventricular output is pushed through the pulmonary outflow, and it also briefly reduces return of desaturated blood from the legs; the knee-chest position works the same way.

· Chronic hypoxemia causes clubbing and secondary polycythemia, and the right-to-left shunt allows paradoxical emboli and brain abscess.

· Fever or a hot bath lowers SVR and worsens cyanosis, and exercise does the same through muscle arteriolar dilation.

Distinctions that matter

· A softer murmur during a spell means less flow through the outflow tract, so the quieter child is the sicker one.

· The VSD carries the shunt but makes no sound; the murmur belongs entirely to the outflow obstruction.

Common misconceptions

· Expecting Eisenmenger syndrome in tetralogy. Pulmonary blood flow is reduced, so the pulmonary arterioles are not exposed to high flow and pressure.

· Explaining squatting by increased venous return alone. The main effect is a rise in systemic vascular resistance that redirects right ventricular output to the lungs.

Where it is taught: Section C05, third paragraph, the tet spell cycle, and the worked example on squatting of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C06-BD1

Brain dump 1 · foundational retrieval: Bicuspid aortic valve and its four complications

A competent response contains

· Bicuspid aortic valve is the most common congenital cardiac anomaly, present in about 1 to 2% of people and more often in males.

· The valve forms from cushion tissue in the outflow tract; two of the three cusps fail to separate, usually the right and left coronary cusps, leaving two unequal leaflets often with a raphe marking the fused commissure.

· Most cases are sporadic but it clusters in families, and some familial cases involve NOTCH1 mutations; it is common in Turner syndrome (45,X) and in coarctation of the aorta.

· Two cusps must fold and crease more than three to open fully, and they open to an eccentric, often domed orifice with turbulent flow.

· Calcific aortic stenosis: repeated mechanical stress injures the leaflet endothelium, then lipid infiltration, chronic inflammation, transformation of valve interstitial cells toward an osteoblast-like phenotype, and dystrophic calcification.

· Significant stenosis typically appears one to two decades earlier than with a three-cusp valve, often between the 40s and 60s instead of the 70s or 80s.

· Aortic regurgitation: a larger fused cusp may prolapse, or a dilated root may pull the commissures apart, so regurgitation is more typical in younger patients and stenosis in older ones.

· Infective endocarditis: damaged leaflet surfaces and turbulent jets favor platelet and fibrin deposits that circulating bacteria can colonize.

· Ascending aortic aneurysm or dissection: intrinsic medial weakness plus eccentric jet stress.

· The auscultatory clue is an early systolic ejection click from the domed valve snapping open, heard while the leaflets are still mobile and lost as they calcify.

Distinctions that matter

· One structural defect explains four complications, so a single patient may develop several over a lifetime.

· The process of calcification is the same as age-related calcific stenosis; the bicuspid valve only runs it faster.

Common misconceptions

· Treating the bicuspid valve as a stenosis problem only. It also leaks, seeds infection, and accompanies an abnormal aorta.

· Expecting the ejection click to persist. It is lost once the leaflets calcify and stop moving.

Where it is taught: Section C06, first three paragraphs and the complications table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C06-BD2

Brain dump 2 · integration and application: Why the aorta is part of the same disease

A competent response contains

· The media shows smooth muscle loss and elastic fiber fragmentation, an intrinsic wall weakness that accompanies the valve defect.

· The eccentric jet from the two-cusp orifice adds focal wall stress, which is the flow contribution.

· Because the medial weakness is intrinsic, aortic dilation can be present even when the valve works normally.

· The dilated ascending aorta can progress to an aneurysm and carries a higher risk of aortic dissection than in the general population.

· Coexisting Turner syndrome or coarctation of the aorta raises that risk further.

· Endocarditis risk is higher because endothelial injury from turbulent flow allows platelet and fibrin deposits to form, which circulating bacteria can then colonize.

· A younger patient is more likely to present with regurgitation from cusp prolapse or root dilation preventing coaptation, while an older patient presents with stenosis after years of dystrophic calcification.

Distinctions that matter

· Intrinsic medial degeneration and jet-induced wall stress are separate contributions, and only the first explains dilation in a normally functioning valve.

· Age sorts the two valve complications: prolapse and root dilation act early, calcification accumulates late.

Common misconceptions

· Treating the aortic dilation as a consequence of stenosis alone. The wall abnormality travels with the valve defect and can precede any valve dysfunction.

· Assuming a normal valve examination rules out aortic disease in these patients.

Where it is taught: Section C06, third and fourth paragraphs and the complications table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-BD1

Brain dump 1 · foundational retrieval: Coarctation: where it sits and what it does to the pulses

A competent response contains

· The narrowing is localized and almost always juxtaductal, at the junction with the ductus arteriosus or ligamentum arteriosum, just distal to the left subclavian artery.

· A leading explanation is that ductal tissue extends into the aortic wall and constricts along with the ductus after birth.

· It is more common in males, occurs in a substantial minority of girls with Turner syndrome, and is accompanied by a bicuspid aortic valve in most patients.

· The narrowing is a high resistance in series with everything downstream, so pressure upstream rises and pressure and flow beyond it are reduced.

· The arms, supplied by vessels arising proximal to the coarctation, are hypertensive.

· Femoral pulses are weak and delayed relative to the radial pulse, which is radiofemoral delay.

· Leg blood pressure is lower than arm pressure, although normally it is slightly higher.

· The legs may be cool or claudicate with exercise.

· Reduced renal perfusion pressure activates the renin-angiotensin-aldosterone system, which adds to the hypertension.

· The left ventricle hypertrophies against the high afterload, and long-standing proximal hypertension raises the risk of heart failure, aortic dissection, and rupture of cerebral berry aneurysms, which are associated with coarctation.

Distinctions that matter

· Hypertension here is both mechanical, from the fixed resistance, and hormonal, from renal hypoperfusion; naming only one leaves the picture incomplete.

· Normally leg pressure is slightly higher than arm pressure, so an equal reading is already abnormal.

Common misconceptions

· Placing the coarctation proximal to the left subclavian artery. It lies just distal to it, which is why the arms are hypertensive.

· Expecting the femoral pulse to be only weak. It is also delayed, and the delay is a specific clue.

Where it is taught: Section C07, first and third paragraphs of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-BD2

Brain dump 2 · integration and application: The newborn who collapses and the teenager who does not

A competent response contains

· The newborn has a severe narrowing, often with arch hypoplasia, historically called infantile or preductal, and the lower body was being perfused by right ventricular blood flowing through the PDA, which is why lower body cyanosis can be present.

· When the ductus closes in the first weeks, perfusion of the lower body and kidneys collapses, producing shock, acidosis and absent femoral pulses; the acidosis is lactic, from tissue hypoperfusion.

· The teenager has a milder narrowing, historically called adult or postductal, and years have allowed collateral vessels to develop, so the presentation is hypertension found on examination rather than collapse.

· Collateral route: subclavian artery above the coarctation, internal thoracic artery, anterior intercostal arteries, posterior intercostal arteries with flow reversed, descending aorta below the coarctation.

· Flow runs backwards in the posterior intercostal arteries.

· Chest film findings: rib notching along the inferior rib margins, usually from about the third rib downward, the figure 3 sign from the indented contour with dilation above and below, and the enlarged tortuous collateral vessels responsible for the notching.

· The first two ribs are spared because their intercostal arteries arise from the costocervical trunk rather than the aorta.

· Notching takes years to appear and is rarely seen in young children.

· The murmur of flow through the narrowing and the collaterals is heard over the back between the scapulae.

Distinctions that matter

· A ductal-dependent lesion deteriorates when the ductus closes, which separates the neonatal presentation from the chronic one entirely.

· Collaterals are protective in the chronic form and absent in the newborn, which is why one presents with shock and the other with a blood pressure reading.

Common misconceptions

· Expecting rib notching in an infant. Collateral enlargement and bone erosion take years.

· Assuming cyanosis rules out coarctation. In the severe neonatal form the lower body is perfused through the ductus and can be cyanotic.

Where it is taught: Section C07, second paragraph, the collateral flow diagram, and the final paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-RT01

Retrieval exercise: Rebuild the collateral route around a coarctation

A competent response contains

· Step 1: subclavian artery above the coarctation.

· Step 2: internal thoracic artery.

· Step 3: anterior intercostal arteries.

· Step 4: posterior intercostal arteries, with flow reversed.

· Step 5: descending aorta below the coarctation.

· The posterior intercostal arteries enlarge and become tortuous under chronically high collateral flow.

· They run in the costal grooves and erode the inferior margins of the ribs, giving rib notching on the chest film, usually from about the third rib downward.

· The first two ribs are spared because their intercostal arteries arise from the costocervical trunk rather than the aorta.

· Notching takes years to develop, so it is rarely seen in a young child.

Distinctions that matter

· The reversed segment is the posterior intercostal arteries, which is the detail that makes the circuit work: blood must travel from the chest wall back into the aorta.

Common misconceptions

· Drawing the collateral route straight from the subclavian artery to the aorta. It travels through the chest wall by way of the internal thoracic and intercostal vessels.

Where it is taught: Section C07, collateral flow diagram and the following paragraph of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C07-CA01

Clinical application: Hypertension in a 16-year-old with weak femoral pulses

A competent response contains

· Recognize: coarctation of the aorta, juxtaductal and just distal to the left subclavian artery, with a bicuspid aortic valve suggested by the early systolic ejection click and right upper sternal border murmur, and Turner syndrome (45,X) suggested by short stature, webbed neck and broad chest.

· Explain: the narrowing is a fixed high resistance in series, so pressure upstream rises and arm pressure is high, while pressure and flow beyond it fall and leg pressure is low; the femoral pulse is weak because flow and pressure downstream are reduced and delayed because the pulse wave reaches the legs partly through longer collateral paths and through the narrowed segment.

· Differentiate: in a PDA with Eisenmenger physiology the legs are desaturated but pulses are bounding with a wide pulse pressure and the fingers stay pink, with no arm to leg pressure gradient of this kind; in severe neonatal coarctation the infant is in shock with absent femoral pulses when the ductus closes, with no collaterals and no rib notching; here the pulses are present but weak and delayed and the pressure gradient is measurable.

· Investigate: rib notching shows enlarged tortuous posterior intercostal arteries eroding the inferior rib margins, which means long-standing collateral flow, and the indented contour with dilation above and below is the figure 3 sign at the coarctation; record blood pressure in both arms and a leg and palpate radial and femoral pulses together to document the gradient and the delay.

· Decide: the renin-angiotensin-aldosterone system is adding to her hypertension, and she is at risk of heart failure, aortic dissection, and rupture of cerebral berry aneurysms.

· Justify: renal perfusion pressure is low beyond the narrowing, so the kidney responds as it would to hypovolemia and activates renin, raising pressure further proximal to the obstruction; dissection risk comes from the high proximal pressure load acting on an aorta whose media already shows smooth muscle loss and elastic fiber fragmentation as part of the bicuspid valve disease, a risk raised further by Turner syndrome.

Distinctions that matter

· The arm to leg gradient with a delayed femoral pulse is what separates coarctation from the other causes of poor lower body perfusion in this chapter.

· The valve lesion and the aortic wall weakness are part of the same disease as the coarctation, not incidental findings.

Common misconceptions

· Reading the hypertension as primary. A measurable arm to leg gradient with a delayed femoral pulse points to a mechanical cause.

· Treating the renin response as inappropriate. The kidney senses low perfusion pressure and responds normally; the response is what makes the hypertension worse.

Where it is taught: Section C07 throughout, with the bicuspid valve and aortic wall changes in C06 and the PDA reversal in C04 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-BD1

Brain dump 1 · foundational retrieval: Draw the parallel circulations of transposition

A competent response contains

· Normally neural crest cells form a pair of conotruncal ridges that grow in a spiral, so the aorticopulmonary septum twists about 180 degrees, connecting the aorta to the left ventricle and the pulmonary trunk to the right ventricle.

· In transposition the septum grows straight instead of spiraling, so the aorta arises from the right ventricle, usually anterior and to the right, and the pulmonary artery arises from the left ventricle.

· Systemic loop: body to right atrium to right ventricle to aorta and back to the body, carrying desaturated blood.

· Pulmonary loop: lungs to left atrium to left ventricle to pulmonary artery and back to the lungs, carrying highly saturated blood.

· Survival requires mixing through a patent foramen ovale or ASD, a PDA, or a VSD.

· Maternal diabetes before conception is a recognized risk factor, and the lesion is more common in males.

· The circulations are parallel rather than in series because each loop returns blood to itself, so the highly saturated pulmonary loop is useless to the tissues while aortic blood is deeply desaturated.

Distinctions that matter

· Tetralogy and transposition are both conotruncal septation defects, but one is a displacement of the outlet septum and the other a failure of the same septum to spiral.

· A high saturation in the pulmonary loop is not a reassuring finding here, because that blood never reaches the tissues.

Common misconceptions

· Describing transposition as a shunt lesion. The problem is the arrangement of the circuits, and a shunt is what keeps the infant alive.

· Expecting the lungs to fix the saturation. The pulmonary loop recirculates its own blood.

Where it is taught: Section C08, first two paragraphs and the parallel circulations flow diagram of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-BD2

Brain dump 2 · integration and application: Why oxygen barely helps and why day three is worse

A competent response contains

· In the fetus this causes little difficulty, because oxygen comes from the placenta and the foramen ovale and ductus are both open.

· After birth, an infant with an intact ventricular septum depends on a foramen ovale or ASD and a PDA; the ductus is the one closing on day three.

· Severe cyanosis appears within hours to days and worsens as the ductus closes, so anything that keeps the ductus open or enlarges the atrial opening improves mixing.

· Supplemental oxygen raises saturation only slightly because it oxygenates blood in the pulmonary loop, which recirculates to the lungs and reaches the aorta only through whatever mixing remains.

· Mixing must go both ways so that volume entering one circuit is returned, otherwise one loop would fill and the other empty.

· An infant with a large VSD has more mixing, so cyanosis is milder, but pulmonary overcirculation brings early heart failure.

· Untreated, most infants die within the first months of life.

· Transposition is the most common cause of cyanosis appearing on the first day of life, while tetralogy of Fallot is the more common cyanotic lesion overall.

· Because the great arteries lie front to back, the upper mediastinal shadow is narrow, and with a rounded heart this gives the egg-on-a-string appearance.

Distinctions that matter

· Most common on the first day against most common overall is a frequent examination distinction between transposition and tetralogy.

· More mixing trades hypoxemia for volume overload, which is why the infant with a VSD looks pinker but fails earlier.

Common misconceptions

· Expecting oxygen to correct the cyanosis. The limitation is anatomic routing, not alveolar oxygenation.

· Treating a closing ductus as unimportant here. It is one of the few mixing sites, so closure makes the infant markedly worse.

Where it is taught: Section C08, third and fourth paragraphs and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C08-CP01

Comparison table: Tetralogy of Fallot against transposition of the great vessels

A competent response contains

· Developmental error: tetralogy from anterosuperior displacement of the infundibular septum; transposition from failure of the aorticopulmonary septum to spiral.

· Associations: tetralogy with 22q11.2 deletion and trisomy 21; transposition with maternal diabetes and male sex.

· Circulation: tetralogy in series with right-to-left flow across the VSD; transposition parallel, requiring mixing.

· Onset of cyanosis: tetralogy variable and set by the degree of outflow obstruction; transposition within hours to days of birth.

· Pulmonary blood flow: reduced in tetralogy; normal or increased in transposition.

· Response to oxygen: little rise in arterial PO2 in either, because in tetralogy the shunted blood bypasses the alveoli and in transposition the oxygenated blood recirculates to the lungs.

· Chest film: boot-shaped heart with dark lung fields in tetralogy; egg on a string with a narrow mediastinum in transposition.

· Survival dependence: transposition with an intact ventricular septum depends on a patent foramen ovale or ASD and a PDA for mixing, while tetralogy does not depend on the ductus unless pulmonary stenosis is severe enough to limit pulmonary flow critically.

Distinctions that matter

· Both are conotruncal defects of outflow tract septation, but they cause cyanosis by different mechanisms: a right-to-left shunt in one, two separated circuits in the other.

· Pulmonary blood flow sorts them quickly: reduced in tetralogy, normal or increased in transposition.

Common misconceptions

· Expecting transposition to have reduced pulmonary flow because the infant is blue. Pulmonary flow is normal or increased; the blood simply recirculates.

· Treating cyanosis in tetralogy as fixed. It varies with outflow obstruction and with systemic vascular resistance.

Where it is taught: Section C08, comparison table, with the tetralogy mechanism in C05 of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C02-Q01

Board-style question: The loud murmur and the quiet one

Correct answer: A. A restrictive defect maintains a large systolic pressure gradient between the ventricles

Why it is right: A restrictive defect limits flow and preserves a large systolic pressure difference between the ventricles, and the high-velocity jet through a small orifice generates the loud harsh holosystolic murmur and the palpable thrill.

B is wrong: Greater volume crossing the septum belongs to the larger defect, which is the second infant, and that infant has the softer murmur, so volume does not explain loudness.

C is wrong: Right ventricular pressure approaching left ventricular pressure describes a large nonrestrictive defect, and it reduces the gradient, which makes a murmur softer rather than louder.

D is wrong: Pulmonary vascular resistance normally falls below systemic resistance in every term infant by this age, so it does not distinguish these two infants or explain the difference in loudness.

A competent response contains

· Murmur loudness reflects the pressure gradient across the defect, not the size of the defect.

· A small restrictive defect keeps left ventricular pressure far above right ventricular pressure throughout systole, so the jet is high velocity and the murmur is loud and harsh with a thrill.

· A large nonrestrictive defect lets ventricular pressures equalize, so the murmur softens while the shunt volume and the clinical burden are greater.

Distinctions that matter

· The loudest VSDs are often the smallest, and the quiet murmur with a loud P2 and failure to thrive is the dangerous one.

Common misconceptions

· Equating murmur loudness with the amount of blood crossing the defect.

Where it is taught: Section C02, small versus large VSD table of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C05-Q02

Board-style question: Squatting during a spell

Correct answer: C. Systemic vascular resistance rises, so a greater share of right ventricular output enters the pulmonary outflow tract

Why it is right: The knee-chest position kinks the femoral arteries and raises systemic vascular resistance, so the aortic exit through the VSD becomes the higher resistance route and a greater share of right ventricular output is pushed through the pulmonary outflow tract, raising arterial saturation.

A is wrong: Pulmonary vascular resistance is not the limiting factor in tetralogy; the obstruction is a fixed infundibular narrowing, and the maneuver does not act on the pulmonary arterioles.

B is wrong: Increased venous return from the legs would raise right ventricular preload without changing how output divides between the two exits, and the maneuver actually reduces return of desaturated blood from the legs briefly.

D is wrong: The infundibular obstruction is structural and does not relax with position; nothing in the maneuver widens the outflow tract.

A competent response contains

· The right ventricle has two exits, the fixed high resistance of the stenotic pulmonary outflow and the VSD into the aorta, whose resistance is systemic vascular resistance.

· Flow divides in inverse proportion to resistance, so raising systemic vascular resistance sends a greater share of right ventricular output to the lungs.

· Squatting and the knee-chest position kink the femoral arteries and raise systemic vascular resistance, and they also briefly reduce return of desaturated blood from the legs.

· The murmur softens during a spell because less blood crosses the obstructed outflow tract.

Distinctions that matter

· The maneuver works on the systemic side of the circuit, not on the obstruction itself, which is fixed.

Common misconceptions

· Explaining the maneuver by increased venous return alone, which would not change how right ventricular output divides between its two exits.

Where it is taught: Section C05, tet spell cycle and the worked example on squatting of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C04-Q03

Board-style question: Blue toes and pink fingers

Correct answer: D. Desaturated pulmonary arterial blood enters the aorta distal to the left subclavian artery

Why it is right: Reversed flow through the ductus delivers desaturated pulmonary arterial blood into the aorta distal to the left subclavian artery, so the descending aorta and the legs receive it while the arch vessels, arising proximal to the ductus, supply the arms with fully saturated blood.

A is wrong: Oxygen extraction occurs in capillaries and lowers venous saturation, not arterial saturation, so it cannot create a difference between arterial readings at the hand and the foot.

B is wrong: The left ventricle does not eject selectively into particular branches; the arch vessels and the descending aorta receive the same ejected blood, and the difference arises only where the ductus adds desaturated blood.

C is wrong: Pulmonary venous blood reaches all systemic arteries through the left ventricle at the same time; sequence of arrival does not produce a sustained saturation gradient.

A competent response contains

· The ductus joins the aorta just distal to the origin of the left subclavian artery.

· With Eisenmenger reversal, desaturated pulmonary arterial blood enters the descending aorta beyond the arch vessels.

· The head and arms are supplied by the brachiocephalic trunk and left subclavian artery, which arise proximal to the ductus and carry undiluted left ventricular blood.

· The result is differential cyanosis with cyanotic clubbed toes and pink fingers.

Distinctions that matter

· Differential cyanosis is an anatomic finding that localizes the communication, not a measure of how severe the hypoxemia is.

Common misconceptions

· Attributing the saturation difference to tissue oxygen extraction. Pulse oximetry reports arterial saturation, which extraction does not change.

Where it is taught: Section C04, final paragraph and the stop and predict box of the CV-18-G Congenital Heart Conditions notes.

NPLEX-P1-CARD-18-07-C03-Q04

Board-style question: A second sound that does not move

Correct answer: B. Right ventricular stroke volume stays nearly constant because inspiratory venous return is offset by reduced shunting

Why it is right: The left-to-right shunt and systemic venous return vary with respiration in opposite directions, so right ventricular stroke volume and the delay in pulmonic valve closure stay nearly constant and the two components of S2 remain equally separated in inspiration and expiration.

A is wrong: Elevated pulmonary arterial pressure would be expected to narrow or alter the split and would come with a low room air saturation if the shunt had reversed, which is not the picture here.

C is wrong: Left ventricular stroke volume is reduced in a large ASD, but an early aortic closure would widen the split variably with respiration rather than fix it, and it does not explain the loss of respiratory variation.

D is wrong: An incomplete right bundle branch block pattern is common in a secundum ASD and contributes to the width of the split, but conduction delay is fixed in both respiratory phases only in appearance and does not explain why the normal inspiratory widening is absent; the respiratory offset of right ventricular volume is what does.

A competent response contains

· Normally inspiration increases systemic venous return to the right heart, delays pulmonic valve closure, and widens the split, which narrows again in expiration.

· In an ASD the right ventricle carries a large stroke volume in both phases of respiration.

· In inspiration the extra systemic venous return is offset by reduced left-to-right shunting, and in expiration the shunt increases again.

· Right ventricular volume, and therefore the timing of P2, stays nearly constant, so the split is wide and fixed.

· The systolic murmur comes from increased flow across a normal pulmonic valve, and normal room air saturation confirms the shunt is still left to right.

Distinctions that matter

· Wide splitting comes from the large right ventricular stroke volume, while fixity comes from the respiratory offset between venous return and shunt flow.

Common misconceptions

· Reading a fixed split as evidence of pulmonary hypertension. Saturation is normal here, so the shunt has not reversed.

Where it is taught: Section C03, paragraph on the fixed split S2 of the CV-18-G Congenital Heart Conditions notes.